

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MENTALCLOUDS | **Context:** South Asia and MENA NLP/AI Practitioner Cohort — MentalCLOUDS Assessment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Reference summaries were produced by augmenting MEMO with annotated dialogue summaries [Q17], with limited documentation of the initial labelers. The qualitative-evaluation expert pool is fully India-affiliated: five healthcare professionals (two clinical psychologists, three psychiatrists/medical practitioners) [Q41, Q42], aged 40–55, four male and one female [Q43], from IIT Delhi, AIIMS Rishikesh, and AIIMS New Delhi [Q9]. There is no documented MENA representation, and web searches confirm that 'no MENA-based mental health professional validation of MentalCLOUDS annotations has been conducted or announced' [WEB-13]. For the HIGH-priority OC dimension, this means salience judgments embedded in reference summaries reflect CBT-aligned Indian/US professional consensus and may not correlate with MENA professional judgments — a convergent-validity concern. The paper documents inter-rater variance ('raters were more aligned in rating MentalBART... with lesser variance' [Q50]; 'fluctuations in how hallucinations are perceived among different models' [Q57]) but does not analyze this variance by professional background or regional training. Aggregation by simple averaging across five raters [Q47] further risks erasing minority perspectives, though the small pool limits the severity of that concern. Notably, the affiliated practitioner organization YourDOST [Q9] strengthens India-specific OC validity for South Asian academic users somewhat but does not extend to MENA contexts.

ID	Evidence
Q9	"Department of Electrical Engineering, Indian Institute of Technology Delhi, India; Department of Computer Science & Engineering, Indraprastha Institute of Information Technology Delhi, India; YourDOST, Karnataka, India; Department of Psychiatry, All India Institute of Medical Sciences, Rishikesh, India; Department of Psychiatry, All India Institute of Medical Sciences, New Delhi, India; Yardi School of Artificial Intelligence, Indian Institute of Technology Delhi, India" (p.1)
Q17	"Expanding upon the MEMO dataset, we augment it with annotated dialogue summaries corresponding to the three identified" (p.6)
Q41	"In order to conduct a comprehensive expert assessment, five healthcare professionals were employed to assess the clinical appropriateness of the summaries produced by the LLMs based on the evaluation framework of Sekhon et al. [39]." (p.12)
Q42	"Among them were two clinical psychologists, with the remaining three comprising psychiatrists and medical practitioners." (p.12)
Q43	"Of the group, four were male, and one was female, with ages ranging from 40 to 55 years and possessing over a decade of therapeutic experience." (p.12)
Q47	"Table 6 reports the clinical experts' scores averaged over the ratings given by five experts." (p.13)
Q50	"Overall, all the raters were more aligned in rating the MentalBART model with lesser variance as compared to the other two LLMs for all the metrics." (p.13)
Q57	"The data shows fluctuations in how hallucinations are perceived among different models and stresses the importance of reviewing evaluations from numerous appraisers for a complete assessment." (p.14)
WEB-13	JMIR Mental Health 2024 — confirms no MENA-based mental health professional validation of MentalCLOUDS annotations conducted or announced; expert pool exclusively India-affiliated (https://mental.jmir.org/2024/1/e57306)

Strengths

- Five expert raters with >10 years therapeutic experience [Q43] provide non-trivial clinical grounding for South Asian academic deployment within Indian institutional clinical environments.
- Inter-rater variance is reported [Q50, Q57], enabling some calibration of label reliability.
- YourDOST industry affiliation [Q9] connects label provenance to active South Asian clinical practice.

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Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Partially. For South Asian academic researchers in CBT-aligned Indian institutional settings, ground-truth labels likely reflect their stakeholder perspective. For MENA practitioners and South Asian practitioners integrating religious or family-system frameworks, the labels are unlikely to fully reflect their perspectives [\[WEB-13, WEB-4\]](#).

OC-2: Disagreement risk is high for MENA stakeholders given (a) zero MENA representation in the annotator pool [\[Q41–Q43, WEB-13\]](#), (b) Islamic psychology frameworks not encoded [\[WEB-4, WEB-5, WEB-6\]](#), (c) per A3, annotators 'did not consciously account for regional context.'

OC-3: Annotator demographics are partially documented: five experts, two clinical psychologists / three psychiatrists/medical practitioners, ages 40–55, 4M/1F, >10 years experience [\[Q41, Q42, Q43\]](#). Institutional affiliations (IIT Delhi, AIIMS Rishikesh, AIIMS New Delhi, YourDOST) are documented [\[Q9\]](#). Initial-labeling annotator demographics for the reference summaries are not specified in the paper [\[Q17\]](#).

OC-4: Re-annotation by a representative regional pool is warranted. No MENA validation exists [\[WEB-13\]](#); ideally a MENA mental-health-professional cohort (Gulf-region clinical psychologists with Islamic-framework training) and a broader South Asian pool (Pakistan, Bangladesh, Sri Lanka) should re-annotate a sample.

OC-5: Aggregation by simple average across five raters [\[Q47\]](#) risks erasure of minority perspectives, though the small pool limits the practical severity. Inter-rater variance is reported [\[Q50\]](#) but not decomposed by professional background or region.

OC-6: Documented label-content issues: (a) zero MENA representation [\[WEB-13\]](#), (b) all-India institutional affiliation [\[Q9\]](#), (c) gender imbalance (4M/1F) [\[Q43\]](#), (d) reference-summary annotator demographics underdocumented [\[Q17\]](#), (e) no decomposition of inter-rater variance by clinical framework or region [\[Q50, Q57\]](#).

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WEB-4	TIIP framework — Islamic psychology clinical taxonomy not represented in the annotator pool's training (https://www.routledge.com/Applying-Islamic-Principles-to-Clinical-Mental-Health-Care-Introducing-Traditional-Islamically-Integrated-Psychotherapy/Keshavarzi-Khan-Ali-Awaad/p/book/9780367488864)
WEB-5	Keshavarzi & Haque four soul aspects framework — additional MENA clinical normative reference (https://www.tandfonline.com/doi/abs/10.1080/10508619.2012.712000)
WEB-6	Ruqyah, dhikr, salah documented as cultural/spiritual treatment adjuncts in Muslim mental health literature (https://pmc.ncbi.nlm.nih.gov/articles/PMC8267770/)
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Information Gaps

- Initial reference-summary annotator demographics for the MEMO expansion [Q17] are not detailed.
- No analysis of inter-rater variance by professional sub-specialty or training tradition.

Requires Expert Verification

- MENA and South Asian (non-India) mental health professionals should re-annotate a sample of MentalCLOUDS reference summaries to quantify regional disagreement.

Remediation

Gap 1: Five-rater India-affiliated expert pool [Q9, Q41–Q43] with no MENA representation [WEB-13]; gender imbalance (4M/1F); aggregation by simple averaging [Q47] without decomposition by clinical framework.

Recommendation: Recruit a parallel MENA + non-India South Asian (Pakistan, Bangladesh, Sri Lanka) expert pool to re-annotate a sample of MentalCLOUDS reference summaries; report inter-rater variance decomposed by region and clinical framework; preserve minority/regional rating distributions rather than averaging.

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